

Linear models lecture

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Introduction

Goals for today

- ▶ Introduce linear regression
- ▶ Learn how to interpret model output
- ▶ Understand diagnostic plots
- ▶ Learn about transformations

History

- ▶ ANOVA, ANCOVA, regression, t test are all variations of the same animal, the *general* linear model
- ▶ Many people (including the R project) call it a linear model (`lm`) to distinguish it from the *generalized* linear model (`glm`)
- ▶ these models are typically fit by ordinary least squares

Extended linear models

- ▶ *Generalized* linear models can incorporate:
 - ▶ (Some) non-linear relationships
 - ▶ Non-normal response *families* (binomial, Poisson, ...)
- ▶ *Mixed* models incorporate *random effects*
 - ▶ Categories in data represent samples from a population
 - ▶ e.g. species, sites, genes ...
 - ▶ Traditionally used to account for experimental blocks

Basic theory

Assumptions

- ▶ *Response variables* are linear functions of *input variables*, in turn based on *predictor variables*
 - ▶ Can have one or more input variables per predictor variable
 - ▶ Each input variable is associated with an estimated parameter (more about this later)
- ▶ *Errors or residuals* are Normally distributed
 - ▶ In other words, the difference between our model *predictions* and our observations is Normal
 - ▶ *not* assuming the marginal distribution is Normal (e.g. a histogram of an input variable)
 - ▶ (Go to board/paper for residual drawing)
- ▶ Predictor variables are independent

Machinery

- ▶ *least squares* fit - we get parameters that minimize the squared differences between predictions and observations
- ▶ Least squares fits have a lot of nice properties
- ▶ Sensitive to some departures from the assumptions
 - ▶ anomalous events tend to have a larger effect than they should

One-parameter variables - continuous

- ▶ Continuous predictor variable: estimate a straight line with one parameter
 - ▶ Also implies one *input variable*
- ▶ $Y = a + bX$
- ▶ Y is the response
- ▶ X is the input variable
- ▶ b is the *slope* and the expected change in Y per unit change in X

One-parameter variables - categorical

- ▶ Categorical predictor variable with two categories: only one parameter
 - ▶ difference in predicted value between levels
- ▶ Parameters are (usually) easy to interpret

You can read in batdat, each row is a bat

```
# in this dataset, each row is a bat that  
# is positive or negative for P. destructans
```

```
batdat=read.csv("bat_data.csv")  
head(batdat)
```

##	swab_id	gd	gdL	swab_type	state	site
## 1	KL15WI0002	1	0.00007560	BAT	WI	HORSESHOE BAY
## 2	KL15WI0003	1	0.47879100	BAT	WI	HORSESHOE BAY
## 3	KL15WI0004	0	NA	BAT	WI	HORSESHOE BAY
## 4	KL15WI0005	1	0.00000551	BAT	WI	HORSESHOE BAY
## 5	KL15WI0006	1	0.00003560	BAT	WI	HORSESHOE BAY
## 6	KL15WI0007	1	0.00003160	BAT	WI	HORSESHOE BAY

##	country	count
## 1	u.s.	3
## 2	u.s.	1110
## 3	u.s.	1110
## 4	u.s.	1110
## 5	u.s.	1110
## 6	u.s.	1110

Some transformations (discussed more later)

```
batdat$lgdL = log10(batdat$gdL)
batdat$count = log10(batdat$count)
```

Running a linear model in R - continuous predictor

```
mod1 = lm(lgdL~lcount, data = batdat)
summary(mod1)
```

```
##
```

```
## Call:
```

```
## lm(formula = lgdL ~ lcount, data = batdat)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -3.0111 -0.9158 -0.1918  0.8661  3.4968
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  -1.6718      0.2507  -6.669 1.75e-10 ***
```

```
## lcount       -0.7043      0.1326  -5.312 2.47e-07 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
```

```
##
```

```
## Residual standard error: 1.257 on 240 degrees of freedom
```

Interpreting model output in R - continuous predictor

- ▶ $y = ax + b$ / log10 counts = $-0.7043x + (-1.6718)$
- ▶ First line - intercept (-1.6718), P-value: probability value is >0
- ▶ Second lines: labeled with parameter names
 - ▶ Predictor is *continuous* so value is slope (-0.704) (or coef)
 - ▶ The value represents the change in the predicted value of the y for each one-unit increase in the x variable (holding all other variables constant)
 - ▶ p-value: when *continuous*, the probability of observing a coefficient as extreme as the one calculated if the true coefficient were actually zero (the null hypothesis)
 - ▶ in plain language: whether your x variable has a significant effect on your y
 - ▶ Std. Error: An estimate of the uncertainty of the coef. estimate.

Running a linear model in R - categorical predictor

```
mod2 = lm(lgdL~species, data = batdat)
summary(mod2)
```

```
##
```

```
## Call:
```

```
## lm(formula = lgdL ~ species, data = batdat)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -3.2199 -0.9052 -0.1714  0.8358  3.2232
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)    -3.64246     0.38222   -9.530  < 2e-16 **
```

```
## speciesMYLU      0.61607     0.39185    1.572  0.116873
```

```
## speciesMYSE     -0.04583     0.71507   -0.064  0.948939
```

```
## speciesPESU      1.60276     0.43789    3.660  0.000294 **
```

```
## speciesSUBSTRATE -0.46844     0.40312   -1.162  0.246072
```

```
##
```

Interpreting categorical data is trickier

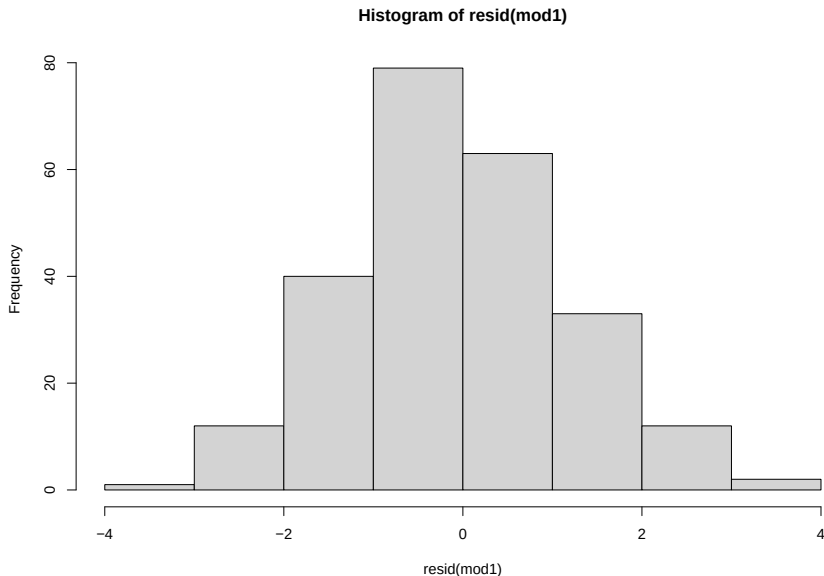
- ▶ First line (labeled INTERCEPT) is actually the REFERENCE
- ▶ This is the omitted group; (EPFU) in this case
- ▶ Each other parameter listed is in REFERENCE to the estimated value of the this variable!
- ▶ e.g. speciesMYLU is saying that MYLU have, on average, 0.61607 more fungus than EPFU
- ▶ you can manually change the reference level of the variables or use other packages to change these contrasts (more on this next week)

Diagnostics

- ▶ Because the linear model is sensitive (sometimes!) to assumptions, it is good to evaluate them
- ▶ Concerns:
 - ▶ *Heteroscedasticity* (does variance change across the data set, check diagnostic plots)
 - ▶ Linearity (does your model fit well?)
 - ▶ Normality (assuming no overall problems, do your **residuals** look Normal?)
 - ▶ Independence (no autocorrelation)
 - ▶ current value is independent of the previous (historic) values - important for time series data
- ▶ Normality is the **least important** of these assumptions

First visualize residuals

```
hist(resid(mod1))
```



- Doesn't look terrible. but can we be certain?

Using qq plots

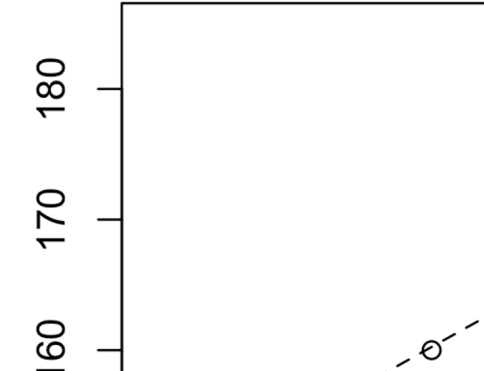
The quantiles
(percentiles) for
the sample data

Sample Quantiles



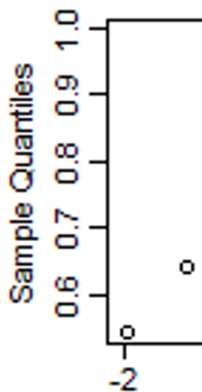
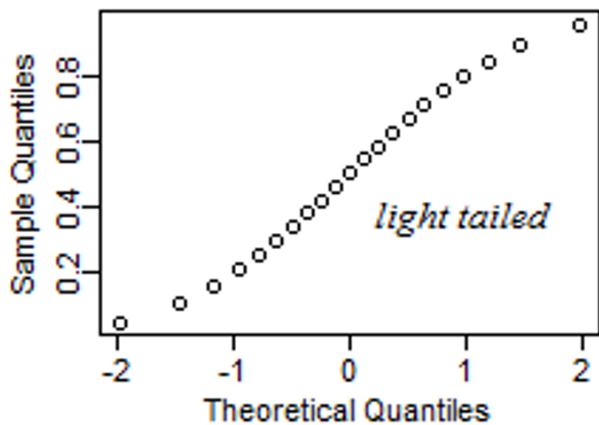
160 170 180

N

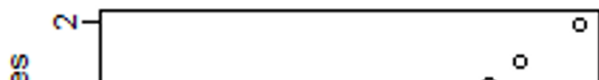


Solving issues

Normal Q-Q Plot

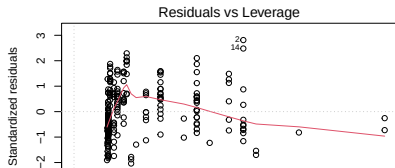
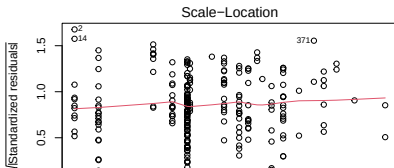
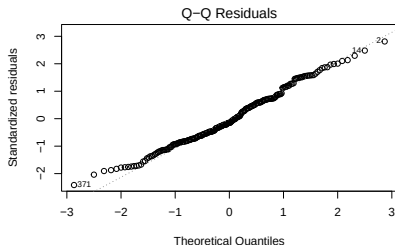
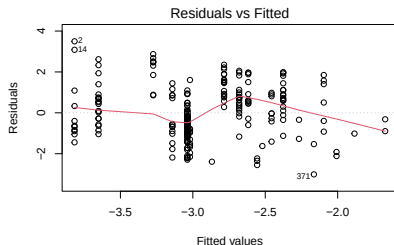


Normal Q-Q Plot



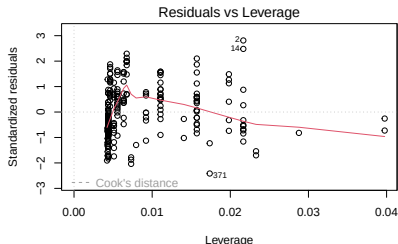
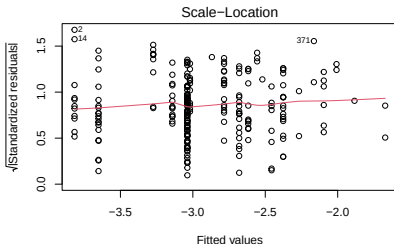
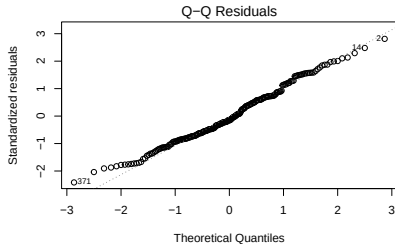
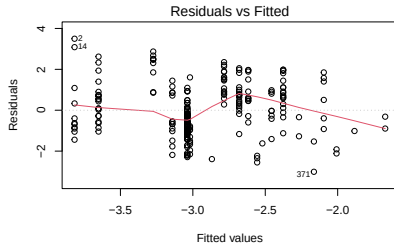
Default plots in R - Heteroscedasticity

- ▶ The plots on the left look at variance across the range of fitted values
- ▶ They ask - as variable one increases, is it's explanatory ability similar across all values? These plots should be flat. Points should be randomly scattered.



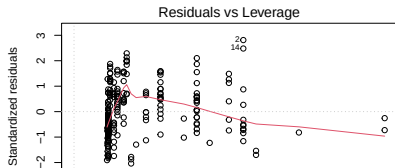
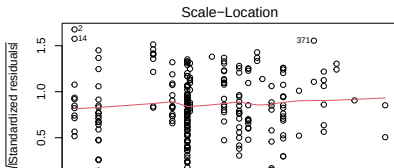
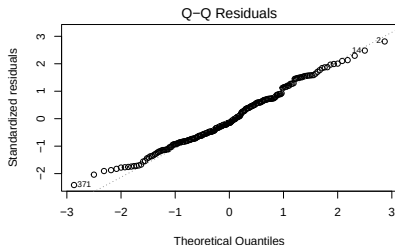
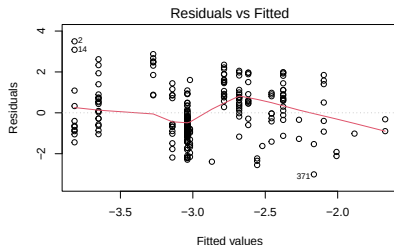
Default plots in R - Normality of Residuals

- ▶ The plot on the top right is a qq plot.
- ▶ It examines normality of residuals - this line should be 1:1.



Default plots in R - leverage of points

- ▶ The plot on the bottom right examines which points have the greatest influence on the regression. I do not recommend removing “real” outlier points but you should examine your results with and without them to see how much they influence your inference.



Using the performance package

- ▶ The performance package is a useful package for doing model checks and improves on some of the plots in base R
- ▶ You can read more about it here:
- ▶ <https://easystats.github.io/performance/>
- ▶ <https://easystats.github.io/see/articles/performance.html>

Basic model diagnostic plots with performance

- ▶ first you will need to install the package and run it

```
#install.packages(performance)
```

```
#load performance
```

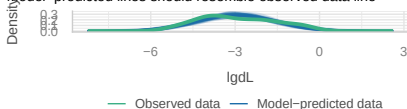
```
library(performance)
```

Diagnostic plots using `check_model` in the performance package

```
check_model(mod1)
```

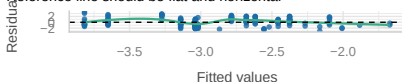
Posterior Predictive Check

Model-predicted lines should resemble observed data line



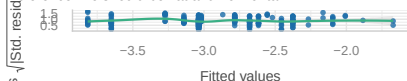
Linearity

Reference line should be flat and horizontal



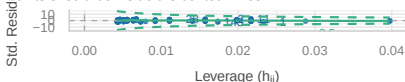
Homogeneity of Variance

Reference line should be flat and horizontal



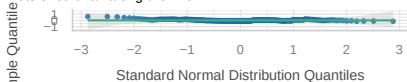
Influential Observations

Points should be inside the contour lines



Normality of Residuals

Dots should fall along the line

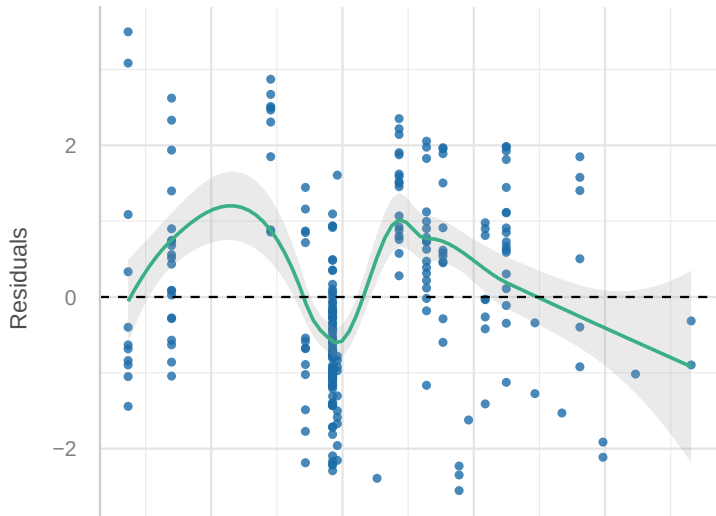


Diagnostic plots using check_model - linearity

```
check_model(mod1, check = "linearity")
```

Linearity

Reference line should be flat and horizontal

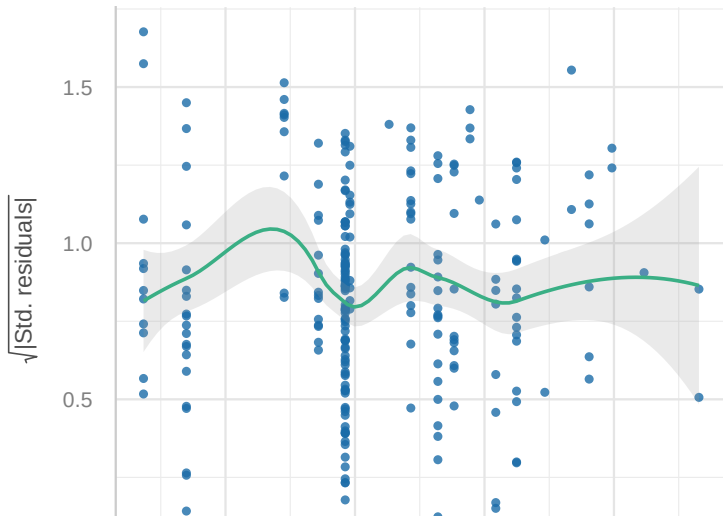


Diagnostic plots using check_model - homogeneity

```
check_model(mod1, check = "homogeneity")
```

Homogeneity of Variance

Reference line should be flat and horizontal

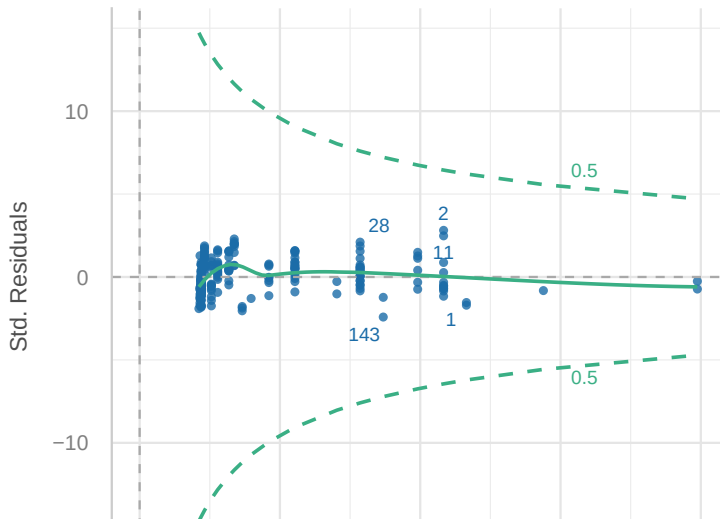


Diagnostic plots using check_model - homogeneity

```
check_model(mod1, check = "outliers")
```

Influential Observations

Points should be inside the contour lines

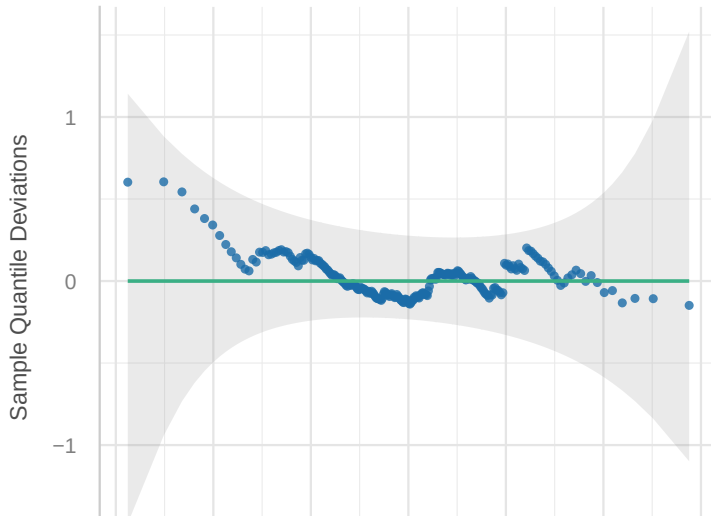


Diagnostic plots using check_model - normality of residuals

```
check_model(mod1, check = "qq")
```

Normality of Residuals

Dots should fall along the line



What do these tell you?

- ▶ The output is similar to base R but the annotation is better!
- ▶ Posterior predictive checks (Bayesian) can be used to “look for systematic discrepancies between real and simulated data by seeing whether simulating the regression coefs result in lines that look like our data
- ▶ Posterior Predictive Check: The predicted data (blue) should roughly match the observed data (black line).
- ▶ Linearity: The green line should be flat and horizontal.
- ▶ Homoscedasticity: Points should be evenly spread, not forming a funnel shape.
- ▶ Q-Q Plot: Residuals should fall along the diagonal dashed line.

Some other assumptions

- ▶ the number of datapoints is greater than the number of predictors
- ▶ there is some variability in the values of your predictor (e.g. x is not the same)
- ▶ your predictors aren't perfectly correlated (e.g. multicollinearity)

Transformations

- ▶ One way to deal with problems in model assumptions is by transforming one or more of your variables
- ▶ Transformations are not cheating: a transformed scale may be as natural (or more natural) a way to think about your data as your original scale
- ▶ The linear scale (no transformation) often has direct meaning, if you are adding things up or scaling them (as in our ant example)
- ▶ The log scale is often the best scale for thinking about physical quantities: 1:10 as 10:?

A variety of transformations exist to normalize skewed data

Skew	Moderate	
Positive (right tail)	Square root transformation	
Negative (left tail)	Reflect then square root transformation	

Transformation tradeoffs

- ▶ A transformation may help you meet model assumptions
 - ▶ Homoscedasticity
 - ▶ Linearity
 - ▶ Normality
- ▶ But there is no guarantee that you can fix them all
- ▶ Piles of zeros are hard too (consider GLMs)

Transformations FAQs

- ▶ Usually we are talking about transforming the Y variable, but transforming the X can also improve model fit
- ▶ Avoid classical 'transform then linear model' recommendations for
 - ▶ probability data or count data
 - ▶ Generally better to respect the structure of these data with a GLM

Tools for fitting and inference

Basic tools

- ▶ `lm` fits a linear model
- ▶ `summary` prints statistics associated with the *parameters* that were fitted

Plotting

- ▶ `plot` can be applied to an `lm` object to give you a nice set of diagnostic tests.
- ▶ `predict` can give predicted values, and standard errors.
- ▶ In `ggplot`, `geom_smooth(method="lm")` fits a linear model to each group of data (i.e. each group that you have identified by plotting it in a different colour, within a different facet, etc.

Assignment - Thursday

- ▶ Make a univariate linear model for one of your hypotheses
- ▶ Examine the assumptions of linearity (using diagnostic plots)
- ▶ Explain what these are telling you (it's okay if they aren't normal, but tell me why)
- ▶ Plot the relationship in ggplot using `stat_smooth` or `stat_summary`